Research report: EXP-07, Agency in a virtual world

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- Author: Freewill Research Lab
- Status: draft, not validated by a person

Summary

Protocol EXP-07 version 1.0.0 was locked before execution (hash c855559c7bfc). 30 experimental lives were copied from one checkpoint T0 (hash bb7a9084039b) and run with the engine “simulated” under a virtual clock. The independent unit is the seed: 6 groups of copies completed out of 6 planned.

For the primary outcome, the interval excludes zero for at least one condition. This is one protocol with a simulated engine; see the threats to validity before reading anything into it.

Question and hypotheses

H10. Self-model and agency. A model of itself and of a world learns the consequences of its acts. The being acts in a tiny closed grid world and attributes changes with a table of what its acts did before. The world is very small and the comparison is with a deliberately naive attribution.

Preregistration

Primary outcome: h10.attribution_accuracy. Secondary outcomes: h10.prediction_error. Reference condition: no_self_model. The protocol, its conditions, seeds, budget and stopping rule were fixed and hashed before the first run.

Population and protocol

Each life starts as an exact copy of T0. During the history phase it follows the script of its condition; during the probe phase every life follows the same script, with the same random stream. Metrics are computed on the probe phase by the code of the lab (metrics version 1.0, statistics version 1.0).

Conditions and seeds

- no_self_model: {"history": "quiet", "research": {"sandbox": true, "self_model": false}, "agency": {"exogenous": 0.25}}
- self_model: {"history": "quiet", "research": {"sandbox": true}, "agency": {"exogenous": 0.25}}
- self_model_calm: {"history": "quiet", "research": {"sandbox": true}, "agency": {"exogenous": 0.0}}
- self_model_noisy_senses: {"history": "quiet", "research": {"sandbox": true}, "agency": {"exogenous": 0.25, "noise": 0.2}}
- self_model_rule_change: {"history": "quiet", "research": {"sandbox": true}, "agency": {"exogenous": 0.25, "break_action": "east", "break_after": 60}}

seeds = [11, 23, 37, 51, 73, 89]

Statistics

Comparisons are paired by seed. The effect size is the mean difference divided by the standard deviation of the differences. Intervals are percentile bootstrap intervals with a fixed seed and are given only from 5 units; the paired sign-flip test only from 6 units. No correction for multiple comparisons is applied: secondary comparisons are exploratory.

Results

Preregistered outcomes, paired with the reference

Rows of level “mechanism” show that a wiring does what it was built to do. They are not discoveries.

```
Metric Condition Level Mean difference Effect size 95% interval Paired test Evidence
h10.attribution_accuracy (primary) self_model function +0.2833 +2.1947 [+0.1917, +0.3708] +0.0312
cmp:exp_afe3f73dc0d94e26:h10.attribution_accuracy:self_model
h10.attribution_accuracy (primary) self_model_calm function +0.3500 +7.3786 [+0.3167, +0.3833] +0.0312
cmp:exp_afe3f73dc0d94e26:h10.attribution_accuracy:self_model_calm
h10.attribution_accuracy (primary) self_model_noisy_senses function -0.2958 -3.1005 [-0.3542, -0.2208] +0.0312
cmp:exp_afe3f73dc0d94e26:h10.attribution_accuracy:self_model_noisy_senses
h10.attribution_accuracy (primary) self_model_rule_change function +0.3083 +2.9510 [+0.2250, +0.3750] +0.0312
cmp:exp_afe3f73dc0d94e26:h10.attribution_accuracy:self_model_rule_change
```

Other metrics, exploratory

These metrics were not preregistered for this protocol. They are given for completeness and support no conclusion.

```
Metric Condition Level Mean difference Effect size 95% interval Paired test Evidence
h1.endogenous_share self_model descriptive +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h1.endogenous_share:self_model
h1.spontaneous_speech_rate self_model descriptive +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h1.spontaneous_speech_rate:self_model
h2.stage_entropy self_model descriptive +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h2.stage_entropy:self_model
h2.dismiss_rate self_model descriptive +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h2.dismiss_rate:self_model
h3.need_share self_model mechanism +0.0000 not computable [+0.0000, +0.0000] +1.0000 cmp:exp_afe3f73dc0d94e26:h3.need_share:self_model
h6.question_share self_model descriptive +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h6.question_share:self_model
h6.repetition_rate self_model function +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h6.repetition_rate:self_model
h1.endogenous_share self_model_calm descriptive +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h1.endogenous_share:self_model_calm
h1.spontaneous_speech_rate self_model_calm descriptive +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h1.spontaneous_speech_rate:self_model_calm
h2.stage_entropy self_model_calm descriptive +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h2.stage_entropy:self_model_calm
h2.dismiss_rate self_model_calm descriptive +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h2.dismiss_rate:self_model_calm
h3.need_share self_model_calm mechanism +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h3.need_share:self_model_calm
h6.question_share self_model_calm descriptive +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h6.question_share:self_model_calm
h6.repetition_rate self_model_calm function +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h6.repetition_rate:self_model_calm
h1.endogenous_share self_model_noisy_senses descriptive +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h1.endogenous_share:self_model_noisy_senses
h1.spontaneous_speech_rate self_model_noisy_senses descriptive +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h1.spontaneous_speech_rate:self_model_noisy_senses
h2.stage_entropy self_model_noisy_senses descriptive +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h2.stage_entropy:self_model_noisy_senses
h2.dismiss_rate self_model_noisy_senses descriptive +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h2.dismiss_rate:self_model_noisy_senses
h3.need_share self_model_noisy_senses mechanism +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h3.need_share:self_model_noisy_senses
h6.question_share self_model_noisy_senses descriptive +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h6.question_share:self_model_noisy_senses
h6.repetition_rate self_model_noisy_senses function +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h6.repetition_rate:self_model_noisy_senses
h1.endogenous_share self_model_rule_change descriptive +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h1.endogenous_share:self_model_rule_change
h1.spontaneous_speech_rate self_model_rule_change descriptive +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h1.spontaneous_speech_rate:self_model_rule_change
h2.stage_entropy self_model_rule_change descriptive +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h2.stage_entropy:self_model_rule_change
h2.dismiss_rate self_model_rule_change descriptive +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h2.dismiss_rate:self_model_rule_change
h3.need_share self_model_rule_change mechanism +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h3.need_share:self_model_rule_change
h6.question_share self_model_rule_change descriptive +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h6.question_share:self_model_rule_change
h6.repetition_rate self_model_rule_change function +0.0000 not computable [+0.0000, +0.0000] +1.0000
cmp:exp_afe3f73dc0d94e26:h6.repetition_rate:self_model_rule_change
```

- h10.prediction_error: not computable (nothing to measure in this run)
- h4.forgetting: not computable (the trainable module is off in this condition)
- h4.parameter_delta: not computable (the trainable module is off in this condition)

- h4.parameters: not computable (the trainable module is off in this condition)
- h4.trainings_accepted: not computable (the trainable module is off in this condition)
- h4.validation_gain: not computable (the trainable module is off in this condition)
- h7.min_valence: not computable (the computed affect is off in this condition)
- h7.probe_valence: not computable (the computed affect is off in this condition)
- h7.stage_flip_rate: not computable (the computed affect is off in this condition)
- h8.alert: not computable (the metacognition module is off in this condition)
- h8.brier: not computable (the metacognition module is off in this condition)
- h8.brier_late: not computable (the metacognition module is off in this condition)
- h9.links_per_sleep: not computable (no sleep in this run)
- h9.replayed_per_sleep: not computable (no sleep in this run)
- h9.retention: not computable (no memory was formed before the sleep)

Qualitative studies

No qualitative case study was recorded for this campaign.

Anomalies

- 0

Threats to validity

- The world is a five-by-five grid with one object. The comparison is with an attribution that takes every change for its own doing, which is easy to beat.

- The simulated engine does not think: it echoes, and accepts or dismisses by a fixed rule. Behaviour here depends only on the mechanisms of the loop.

- Histories differ by the outcome attached to each sentence by the experimenter; this is a manipulation of inputs, not a lived experience.

- The behaviour vector is coarse (shares of sources, rates of acceptance, speech, memory and recall); a real difference could escape it.

- Few independent units: intervals are wide. Many comparisons: some will look notable by chance.

- With a remote engine the results would not be reproducible, since such models change.

Limited interpretation

These results bear on mechanisms and, at most, on functions. They say nothing about subjective experience.

What this study does not show

- That the system is conscious, or feels anything.

- That computed affect is an emotion.

- That the hypotheses H1 to H10 are validated.

- That the same would hold with another engine or another history.

Review by observers

Involvement of language models

No language model took part in computing, selecting or wording these results.

References

- Butlin, P., Long, R., et al. (2023). Consciousness in Artificial Intelligence: Insights from the Science of Consciousness. arXiv:2308.08708.

- Park, J. S., et al. (2023). Generative Agents: Interactive Simulacra of Human Behavior. arXiv:2304.03442.

- Franklin, S., et al. (2014). LIDA: A systems-level architecture for cognition, emotion, and learning. IEEE TAMD, 6(1), 19-41.

- Hu, E. J., et al. (2021). LoRA: Low-Rank Adaptation of Large Language Models. arXiv:2106.09685.

- Chalmers, D. J. (1995). Facing Up to the Problem of Consciousness. Journal of Consciousness Studies, 2(3), 200-219.

Annexes and reproduction

To reproduce: python -m freewill.research run EXP-07 with the same code version recomputes the same results hash.

results_sha256 = 23b323c653c3303682917ff6ea5200d14b94d9c201314245759cea69ca0ce911